

Igor Kupchenenko

Backend Software Engineer · Go · Distributed Systems & Data Pipelines

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SUMMARY

Backend software engineer with 3+ years of professional Go experience designing and shipping distributed systems in high-throughput production environments. Built event-driven microservices with Kafka and Protobuf, decomposed monoliths into service-oriented architectures using Domain-Driven Design and Event Sourcing, and profiled high-load services from seconds to milliseconds. Strong SQL and data foundation — advanced querying, indexing, and data modeling — with hands-on data-warehouse work: building Airflow batch-ingestion pipelines and analytical reporting on payment-gateway data. Comfortable owning services end to end and working across teams to align contracts and ship reliably.

English: B2

SKILLS

Languages: Go (primary), Python, SQL

Data & SQL: PostgreSQL, advanced SQL (window functions, joins & set operations, indexing — clustered vs non-clustered, query optimization), ACID, transactions, data modeling

Data Pipelines & Warehousing: Apache Airflow (batch ingestion, DAG orchestration), ETL, data-warehouse (DWH) reporting, analytical reporting

Architecture & Design: Microservices, Domain-Driven Design, Event Sourcing, Clean Architecture, RESTful & gRPC API design, system design

Messaging & Streaming: Kafka, RabbitMQ, Protobuf, Redis (caching, pub/sub), webhooks

Infrastructure & Cloud: Docker, Kubernetes, AWS, Linux, Git, CI/CD pipelines

Observability: Prometheus, Grafana, structured logging, distributed tracing

Testing: Unit, integration, table-driven tests, benchmarks

Ways of working: Agile / Scrum, code review, cross-functional collaboration

EXPERIENCE

Backend Engineer · Paycos

07/2024 – 12/2025

- Designed and built the Order Gateway microservice as a validation and aggregation layer, unifying 5+ heterogeneous data sources and ensuring transaction consistency across the platform.
- Built a Callback Service consuming Kafka events and delivering real-time webhook notifications to 50+ merchant integrations; defined event schemas, retry semantics, and idempotency.
- Migrated Kafka models and Redis caching layers from JSON to Protobuf, reducing service latency by ~45% and doubling throughput.
- Built an Expiration Service to automatically poll and expire stale orders, eliminating manual intervention and improving uptime by 12%.
- Created gRPC authorization middleware with JWT (go-jwx), standardizing token validation across 8 microservices.
- Coordinated API contract design with frontend and product teams, shortening integration cycles by ~2 weeks.
- Maintained 80%+ test coverage across gateway and callback pipelines.

Stack: Go, gRPC, PostgreSQL, Redis, Kafka, Protobuf, Docker, AWS

Backend Engineer · BWG

06/2023 – 07/2024

- Operated within a 21-service Go architecture built on Domain-Driven Design and Event Sourcing.
- Decomposed a monolithic codebase into 6 core services and 40+ peer-to-peer integrations, reducing system latency by 25.3% and enabling independent deployments.
- Profiled and optimized a high-load service from 9.5s to 650ms (93% reduction) using pprof, goroutine tuning, connection pooling, and SQL query optimization.
- Contributed to the data warehouse (DWH): designed and built Airflow batch-ingestion pipelines feeding daily reporting on order completion across payment-gateway systems.

- Delivered analytical reporting on transaction success rates across payment gateways, supporting operational monitoring and decision-making.
- Built a receipt-validation pipeline that flagged ~15% of submissions as fraudulent, preventing an estimated \$30K/month in losses.
- Designed a local OCR pipeline with Tesseract, automating processing of ~2,000 scanned documents/week and replacing manual data entry.
- Reduced maintenance costs by ~50% through codebase cleanup and infrastructure consolidation.

Stack: Go, PostgreSQL, Apache Airflow, Docker, Linux, Git

Backend Engineer • PizzaSushiWok

01/2022 – 06/2023

- Owned the Go backend for the main ordering platform in a 2-person team — full responsibility for design, implementation, and production operation.
- Built a cross-platform mobile app with Flutter; gathered feature requirements with stakeholders and translated them into implementation.
- Built and maintained a Python-based delivery-staff service end to end.
- Refactored core backend components to cut response times by 30% under peak traffic via Redis caching and SQL query optimization.
- Built a Go alerting bot monitoring deployments, errors, and noisy logs, cutting incident resolution time by 30%.

Stack: Go, Python, Flutter, PostgreSQL, Redis, Kafka, RabbitMQ, Docker, Git

PROJECTS

Game Telemetry Pipeline • personal project

in progress

- Building an end-to-end telemetry pipeline for a game in development: event ingestion, storage, transformation, and an analytics layer — driven by the need to understand player behavior. Covers ingestion, data modeling, and transformation.

EDUCATION

Bachelor of Computer & Information Science — Moscow Finance and Law Academy (MFUA), 2019 – 2023

ADDITIONAL

- 6 months tutoring programming and CS fundamentals — comfortable explaining technical concepts to non-technical learners.
- Active interest in data engineering, observability tooling, and developer experience.